

bKash presents NSUCEC Datathon

Predicting FictiPay Customer Churn

A survival-style model of account silence · scored on AUC-ROC

0.98535

OOF AUC

0.9173

PRECISION@10%

0.7235

RECALL@10%

194

FEATURES

The problem, framed as silence

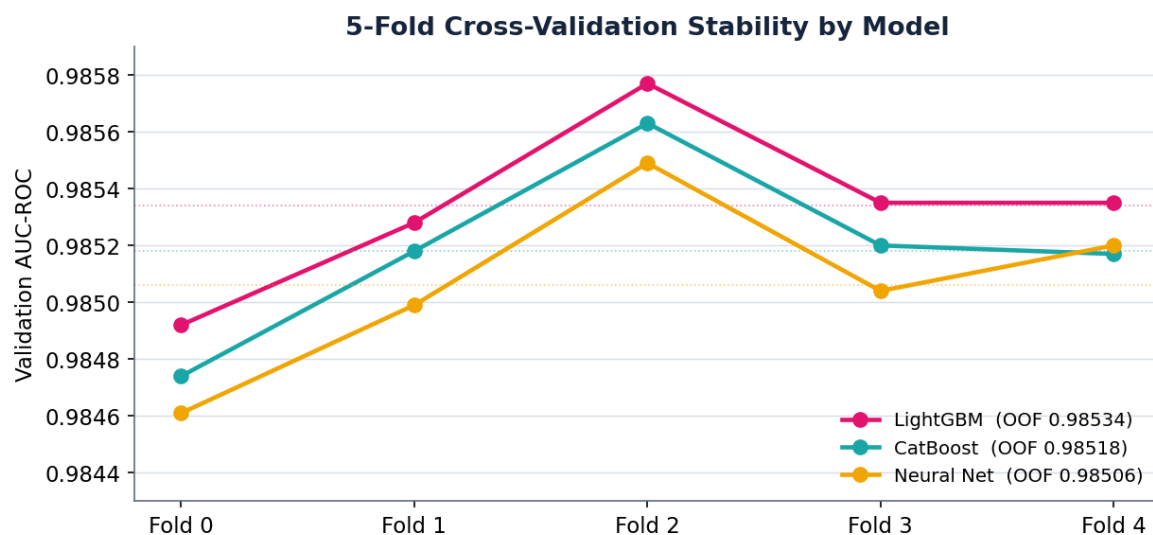
- **Target.**
Predict whether a FictiPay account churns at the 2024-03-31 snapshot, scored by AUC-ROC on a probability in [0,1].
- **Three joined sources.**
Transaction ledger (sent/received), day-end balances, and KYC profile across 595k train accounts.
- **Core idea.**
Churn = the account goes quiet. Model how long it has been silent versus its own normal transaction cadence.
- **194 features in 10 families.**
Recency & windows, Poisson zero-probability, gap-survival, periodicity, trend/decay, balance dynamics, type-mix, KYC.

The three signals that carry the model

Recency days since last sent transaction	0.968
Recent volume sent count in last 21-30 days	0.975
Poisson p(0) prob. of zero future events	0.975
Cadence break silence vs account's own max gap	0.934
Periodicity overdue vs expected next txn (v7)	0.879

Values = standalone single-feature AUC (from notebook diagnostics)

Results — three models, one fair blend



0.98535

FINAL OOF AUC

0.9173

PRECISION @ TOP 10%

0.7235

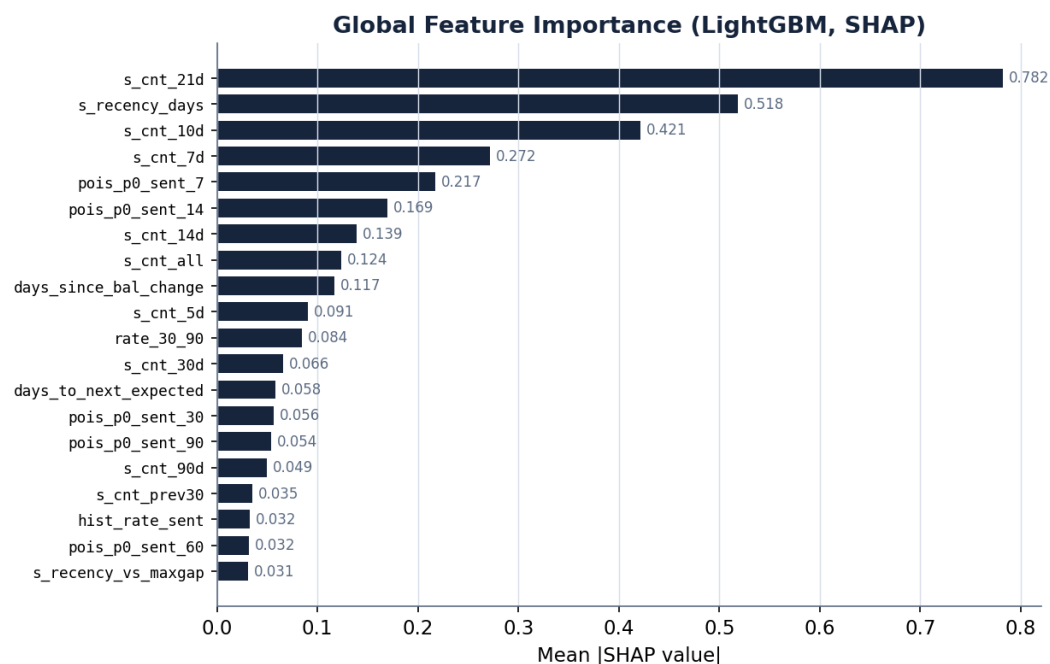
RECALL @ TOP 10%

0.98534

BEST SINGLE (LIGHTGBM)

Weight-optimised blend (lgb 0.745 / cb 0.115 / nn 0.140) selected automatically on OOF — a marginal, honest +0.00002 gain.

What the model actually uses



Reading the drivers

- **Recent activity counts lead.**
21-, 10- and 7-day sent-transaction counts top the SHAP ranking.
- **Recency is the backbone.**
s_recency_days is the 2nd most important feature overall.
- **Balance staleness corroborates.**
days_since_bal_change adds an independent dormancy signal.
- **Periodicity helps the edges.**
days_to_next_expected flags accounts past their due date.

Global feature importance: mean |SHAP| from the final LightGBM fold model on a 20,000-row sample.

Honest takeaways & next step

- **Strong, stable model.**

OOF AUC 0.98535 with tight per-fold spread; captures ~72% of churners in the top 10% of risk at ~92% precision.

- **Ensemble gain is tiny.**

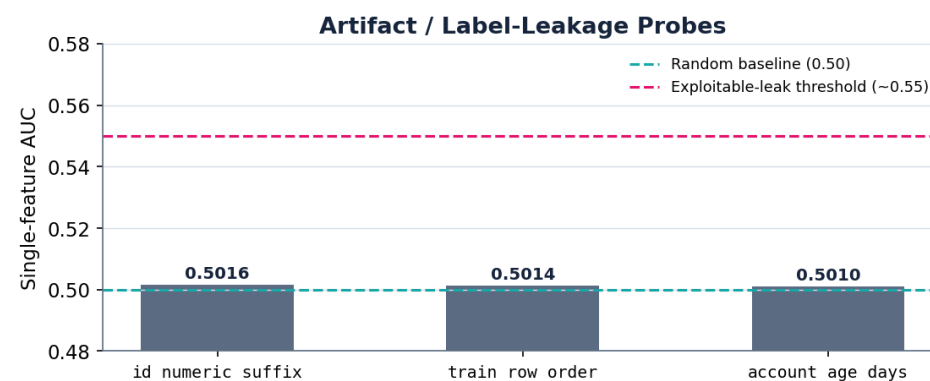
The blend beats the best single LightGBM by +0.00002 — within noise. Kept because it does not hurt.

- **No data leak to exploit.**

ID-order, row-order, and account-age probes all sit at ~0.50 (max 0.5016), so the score is real behavioural signal.

- **0.99 is not supported here.**

Single-feature power plateaus at ~0.975; the residual ~0.014 gap looks like irreducible noise, not missing modelling.



Recommended next step

Treat AUC ~0.985 as the behavioural ceiling for this data. If an F1 number is required, add a separate probability-threshold tuning step — it changes the reported metric, not the underlying model quality.